

Software Requirements Specification

BiletFlow: Self-Service Event Ticketing Platform for Kazakhstan

Version: 0.3

Status: Initial Draft

Project Type: Web and mobile event and ticket management platform

Executive Summary

BiletFlow is a Kazakhstan-focused self-service platform that allows individuals, student organizations, businesses, and community groups to create events and issue digital tickets.

Creating an event and distributing free tickets is free. Organizers who want to sell paid tickets must activate paid sales for the event by paying a one-time activation fee and completing identity and payout verification.

The platform provides event pages, ticket inventory management, online registration, secure payments, QR-code tickets, calendar export, attendee management, and a mobile ticket-verification app for event administrators.

1. Introduction

1.1 Purpose

This document defines the initial requirements for BiletFlow. It describes the platform's users, business model, main features, constraints, and MVP scope.

1.2 Scope

The platform will allow organizers to:

- Create and publish public or private physical events held at a venue.
- Create multiple ticket types.
- Configure assigned-seating events and interactive seat maps.
- Issue free tickets without paying a platform fee.
- Activate paid ticket sales for individual events.
- Set prices, quantities, and sale periods.
- Track registrations and revenue.
- Send digital tickets containing unique QR codes.
- Allow attendees to download and print their tickets.

- Assign event administrators who verify tickets and check attendees in through a mobile app.
- Allow attendees to export event details to their preferred calendars.
- Cancel orders and process refunds where permitted.

Attendees will be able to discover events, register or purchase tickets, receive digital tickets, add event details to their calendars, and present their tickets for check-in.

1.3 Definitions

- Organizer: A user who creates and manages an event.
- Attendee: A person who registers for or purchases a ticket.
- Event Admin: An organizer-authorized user who uses the mobile app to verify tickets and check in attendees. This role is separate from the Platform Admin.
- Free Ticket: A ticket with no payment requirement.
- Paid Ticket: A ticket that requires payment during checkout.
- Paid Sales Activation: The process that enables an organizer to collect money for an event.
- Support Case: A support request linked to a user, event, order, or ticket and represented through an asynchronous chat-like conversation.
- Promo Code: An organizer-created code that applies a configured ticket discount and attributes the resulting order to a campaign.
- Campaign QR Code: A promotional QR code that opens an event page through a trackable link and automatically applies its associated promo code. It is not an admission ticket.
- Platform Admin: Internal staff responsible for moderation, disputes, and platform management.
- Payment Processor: An external service that processes payments and organizer payouts.

2. Overall Description

2.1 Product Perspective

BiletFlow will consist of:

- A responsive web application for organizers and attendees.
- A dedicated iOS and Android application for event administrators to verify tickets and manage check-in.
- An administrative web portal.
- Backend services for events, orders, payments, tickets, support cases, promotional campaigns, notifications, and reporting.
- Integrations with a payment processor, email provider, and calendar-export services.

2.2 Stakeholders

- Event organizers
- Attendees
- Event administrators and check-in staff
- Platform administrators
- Payment and payout providers
- Development and QA teams
- Project sponsors and business stakeholders

2.3 User Classes

Attendee

Can browse events, register, purchase tickets, download tickets, and review existing orders.

Organizer

Can create events, configure ticket types, activate paid sales, manage attendees, view sales, and assign check-in staff.

Event Admin

Can use the mobile app to select assigned events, scan and verify QR-code tickets, search for attendees, and record or reverse check-ins.

Platform Admin

Can moderate events, manage users, review payment activation requests, handle disputes, and view platform reports.

3. Business and Monetization Model

3.1 Free Events

Organizers may:

- Create and publish events without charge.
- Create free ticket types.
- Issue QR-code tickets.
- Use attendee lists and check-in functionality.

- Send basic confirmation messages.

3.2 Paid Events

Paid ticket sales remain disabled until the organizer:

1. Creates an event and at least one paid ticket type.
2. Completes identity and payout verification.
3. Pays the event's Paid Sales Activation Fee.
4. Connects or registers a valid payout account.
5. Accepts the paid-event terms.

Once activated, the organizer can collect ticket payments for that event.

For the academic MVP, identity checks, activation-fee payment, ticket payment, and organizer payouts may be implemented with a provider sandbox or clearly labelled simulation. No real money movement is required for the term demonstration.

3.3 Recommended Initial Pricing Model

- Free events: no platform fee.
- Paid Sales Activation Fee: one-time fee per event, charged in KZT; amount to be determined.
- Payment-processing charges: deducted from each transaction.
- Optional per-ticket service fee: excluded from the initial MVP and considered for a later release.

This model gives organizers the complete basic workflow at no cost and charges them only when they intend to generate revenue.

4. System Features and Requirements

4.1 Account Management

- Users shall be able to register using an email address.
- Users shall be able to verify their email addresses.
- Users shall be able to sign in, sign out, and reset passwords.
- Organizers shall have a profile containing contact and payout information.
- The system shall support role-based permissions.

4.2 Event Management

Organizers shall be able to:

- Create, edit, duplicate, publish, unpublish, and cancel events.
- Add an event title, description, category, images, venue, date, and time.
- Create physical, venue-based events.
- Mark events as public, unlisted, or private.
- Define registration opening and closing times.
- Set an event capacity.
- Preview an event before publication.

4.3 Ticket Management

Organizers shall be able to:

- Create free and paid ticket types.
- Set ticket name, description, price, and quantity.
- Configure ticket sales start and end times.
- Limit the number of tickets purchased in one order.
- Hide ticket types without deleting them.
- View the number of available, reserved, sold, refunded, and checked-in tickets.

The system shall prevent ticket sales when inventory has been exhausted.

4.3.1 Assigned Seating and Seat Selection

- Organizers shall be able to mark an event as general admission or assigned seating.
- The academic MVP shall provide at least one predefined venue layout containing sections, rows, seat numbers, accessible seats, and price categories.
- Organizers shall be able to select and configure the predefined layout; a visual venue-layout editor is not required.
- For assigned-seating events, attendees shall see an interactive seat map during registration or checkout.
- The seat map shall clearly distinguish available, selected, temporarily held, sold, unavailable, and accessible seats with both colors and text or symbols.
- Attendees shall be able to select one or more available seats and see the ticket price before continuing to checkout.
- Selected seats shall be held temporarily while checkout is in progress.
- A seat hold shall expire and release the seat when checkout is abandoned or its time limit is reached.

- The system shall prevent two orders from purchasing the same seat, including when multiple attendees check out concurrently.
- The assigned section, row, and seat number shall be stored on the ticket and order item.

4.4 Free Registration

- An attendee shall be able to reserve a free ticket without entering payment information.
- The system shall create an order with a zero-value total.
- The attendee shall receive a confirmation email and QR-code ticket.
- The organizer shall see the registration in the attendee list.

4.5 Paid Sales Activation

- Paid tickets shall not be purchasable before activation.
- The organizer shall see an activation checklist.
- The system shall record payment of the activation fee.
- The system shall verify that the simulated or sandbox payout requirements are complete.
- Activation shall apply only to the selected event.
- Platform administrators shall be able to suspend paid sales when fraud or policy violations are suspected.

4.6 Checkout and Payments

- Attendees shall be able to select tickets and provide attendee details.
- Ticket inventory shall be temporarily reserved during checkout.
- The system shall calculate ticket totals and processing charges.
- Payments shall be processed through an external provider sandbox or a clearly labelled internal simulation for the academic MVP.
- Tickets shall only be issued after successful payment confirmation.
- Failed or abandoned transactions shall not create valid tickets.
- Organizers shall be able to view gross sales, fees, refunds, and estimated payouts.
- Demonstration payment records shall never be presented as real financial transactions.

4.7 Digital Tickets

Each issued ticket shall include:

- A unique ticket identifier.
- Event and ticket-type information.

- Attendee information.
- A tamper-resistant QR code.
- Current status: valid, checked in, cancelled, or refunded.

Tickets shall be available through email and the attendee's account.

- Attendees shall be able to download each ticket as a print-optimized PDF and print it from a standard browser or PDF viewer.
- A printed ticket shall include the event name, date and time, venue, ticket type, attendee name, assigned section/row/seat when applicable, ticket identifier, and QR code.
- Payment-card details and other unnecessary sensitive information shall not appear on the printed ticket.
- The QR code shall remain large and clear enough to be scanned from a paper copy, including when printed in grayscale.
- Digital and printed copies of the same ticket shall share one ticket identifier and shall not create separate admissions.

4.8 Mobile Ticket Verification and Event Check-In

The system shall provide a dedicated mobile app for authorized event administrators. The app shall allow them to:

- Sign in securely and view only assigned events.
- Scan ticket QR codes using the device camera.
- Validate ticket authenticity and status in real time.
- Display a clear valid, invalid, cancelled, refunded, or already-used result.
- Record a successful check-in and prevent the same ticket from being used twice.
- Search for attendees manually.
- Undo an accidental check-in where authorized.
- View the total number of registered and checked-in attendees.
- Synchronize check-in records with the central platform.

The academic MVP requires online verification only. Offline synchronization is deferred.

4.9 Orders, Cancellations, and Refunds

- Attendees shall be able to view their orders.
- Organizers shall be able to define a refund policy.
- Organizers shall be able to cancel free registrations.
- Authorized organizers shall be able to initiate full refunds.
- Refunded or cancelled tickets shall become invalid.
- All payment and refund actions shall be recorded in an audit log.

4.10 Notifications

The system shall send notifications for:

- Account verification
- Registration or purchase confirmation
- Payment failure
- Ticket delivery
- Event updates
- Event cancellation
- Refund completion
- Organizer payout status
- New support message, support-case assignment, and support-case status changes

4.11 Calendar Export

- Attendees shall be able to download event details as a standard iCalendar (.ics) file.
- The event page and order confirmation shall provide one-click options for common calendar services.
- Calendar entries shall include the event title, start and end times, time zone, venue name and address, description, and event URL.
- Calendar exports shall use a stable event identifier so updated event details can replace an earlier calendar entry.
- When an event is cancelled, the system shall provide an updated cancellation calendar file or link.
- Calendar export shall not require the platform to request write access to an attendee's calendar account.

4.12 Administration

Platform administrators shall be able to:

- Search for users, events, orders, and payments.
- Suspend users or events.
- Review reported events.
- Inspect paid-sales activation records.
- Monitor refunds, disputes, and payment failures.
- Configure activation fees and platform settings.
- Review escalated support cases and promotional-code activity.
- Export basic operational reports.

4.13 Support Chat and Case Management

- An attendee shall be able to open a support request from an event, order, or ticket page.
- An organizer shall be able to open a platform-support request concerning account, activation, payment, or technical issues.
- A support case shall automatically include the relevant user, event, order, and ticket context when available.
- The requester shall select an issue category such as ticket delivery, payment, refund, seating, event information, check-in, account, or technical problem.
- Each support case shall provide an asynchronous chat-like message thread.
- Attendees and organizer support staff shall be able to exchange messages within attendee support cases.
- Organizers and Platform Admins shall be able to exchange messages within organizer support cases.
- Authorized staff shall be able to assign a case and change its status among Open, In Progress, Waiting for Customer, and Resolved.
- Users shall receive in-app or email notifications when a new reply or relevant status change occurs.
- The system shall retain the case history and prevent users from viewing cases they are not authorized to access.
- The academic MVP does not require typing indicators, presence information, voice messages, AI chatbots, or real-time WebSocket delivery.

4.14 Promo Codes and Campaign QR Codes

- Organizers shall be able to create a promotional campaign for an event.
- A campaign may have a percentage or fixed-KZT discount, validity dates, maximum redemptions, and applicable ticket types.
- The system shall generate a unique promo code and a special Campaign QR Code for the campaign.
- The Campaign QR Code shall encode a trackable HTTPS event link containing an opaque campaign or promo token rather than a discount amount trusted by the client.
- Scanning the Campaign QR Code shall open the correct event page and automatically apply the associated promo code after server-side validation.
- The attendee shall see the applied code, discount, and updated order total before completing checkout.
- Expired, disabled, exhausted, or inapplicable codes shall be rejected with a clear message.
- The server shall calculate and validate all discounts and shall prevent redemption beyond the campaign limit.
- The Campaign QR Code shall be visually and functionally distinct from an admission-ticket QR code.

- The mobile ticket-verification app and admission endpoint shall never accept a Campaign QR Code as permission to enter an event.
- BiletFlow shall record exact redemptions, orders, tickets sold, gross revenue, discount amount, and net revenue for each campaign.
- GA4 may record campaign-link visits, checkout starts, and traffic attribution without receiving attendee names, email addresses, phone numbers, ticket identifiers, or other direct personal information.

4.15 Basic Organizer Analytics

- Each organizer shall have access only to analytics for events they manage.
- The dashboard shall show total ticket capacity, tickets sold, tickets remaining, and percentage sold.
- The dashboard shall show gross sales, discounts, refunds, and net demonstration revenue in KZT.
- Organizers shall be able to view sales over time using order timestamps.
- Organizers shall be able to compare sales by ticket type.
- The dashboard shall show campaign and promo-code redemptions, tickets attributed to each campaign, and resulting revenue.
- The dashboard shall show checked-in attendees, absent ticket holders, and the check-in percentage after or during the event.
- Basic filters shall include event, date range, and ticket type.
- Required analytics shall be calculated from authoritative BiletFlow order, ticket, campaign, refund, and check-in records.
- Basic analytics shall not require attendees to provide demographic information or complete additional checkout fields.
- GA4 traffic, funnel, geographic, device, or demographic reporting is an optional advanced analytics feature and is not required for the base MVP.

4.16 Organizer Event History and Audit Trail

- The organizer dashboard shall classify events as Upcoming, Active, Completed, or Cancelled.
- Organizers shall be able to open completed and cancelled events and review their retained tickets, orders, refunds, check-ins, promo campaigns, support cases, and basic analytics.
- Organizers shall be able to duplicate a past event into a new draft while excluding the original event's orders, tickets, payments, check-ins, and support cases.
- Each event shall provide a chronological activity timeline.
- The timeline shall include important actions such as event publication or cancellation, ticket price or capacity changes, refunds, promo-code creation or disabling, support-case status changes, check-ins, and check-in reversals.

- Each history entry shall contain a timestamp, acting user, action type, affected entity, and short description.
- Organizers shall be able to filter history by date range and activity type.
- Organizer staff shall only see history for events they are authorized to manage; Platform Admins may view history for moderation or support purposes.
- Audit entries shall not be editable or deletable through normal organizer interfaces.
- Full field-by-field version history, rollback, and permanent compliance archival are not required for the academic MVP.

5. Example Use Cases

UC1: Organizer Creates a Free Event

Actor: Organizer

Precondition: Organizer has a verified account.

Flow: The organizer enters event details, creates a free ticket type, previews the event, and publishes it.

Postcondition: Attendees can register without payment.

UC2: Organizer Activates Paid Ticket Sales

Actor: Organizer

Precondition: A draft event contains a paid ticket type.

Flow: The organizer completes verification, connects a payout account, pays the activation fee, and accepts the terms.

Postcondition: Paid checkout becomes available for the event.

UC3: Attendee Purchases a Ticket

Actor: Attendee

Precondition: Paid sales are active and tickets are available.

Flow: The attendee selects tickets, enters required information, and completes payment.

Postcondition: An order and QR-code ticket are issued.

UC4: Event Admin Verifies and Checks In an Attendee

Actor: Event Admin

Precondition: The Event Admin is signed into the mobile app and assigned to the event.

Flow: The Event Admin selects the event and scans the attendee's QR code. The mobile app

validates the ticket and records the check-in.

Postcondition: The ticket cannot be checked in again unless the previous action is reversed.

UC5: Organizer Refunds an Order

Actor: Organizer

Precondition: The order is eligible for a refund.

Flow: The organizer selects the order, confirms the refund, and the payment provider processes it.

Postcondition: The ticket is invalidated and the attendee is notified.

UC6: Attendee Adds an Event to a Calendar

Actor: Attendee

Precondition: The attendee is viewing an event page, confirmation page, or issued ticket.

Flow: The attendee selects a calendar option or downloads the .ics file.

Postcondition: A calendar entry is prepared with the correct event time zone, venue name and address, and event URL.

UC7: Attendee Prints a Ticket

Actor: Attendee

Precondition: A valid ticket has been issued.

Flow: The attendee opens the order or ticket, downloads the print-optimized PDF, and prints it.

Postcondition: The paper ticket contains a readable QR code that the Event Admin mobile app can verify at the entrance.

UC8: Attendee Selects an Assigned Seat

Actor: Attendee

Precondition: The event uses assigned seating and sales are open.

Flow: The attendee opens the interactive seat map, reviews seat availability and prices, selects an available seat, and completes checkout before the temporary hold expires.

Postcondition: The purchased seat is linked to the order and ticket and is no longer available to other attendees.

UC9: Attendee Requests Support

Actor: Attendee

Precondition: The attendee is viewing an event, order, or ticket.

Flow: The attendee selects support, chooses an issue category, writes a message, and submits the case. The organizer replies through the support conversation.

Postcondition: The conversation and its event/order/ticket context are retained until the case is resolved.

UC10: Attendee Purchases Through a Campaign QR Code

Actor: Attendee

Precondition: The organizer has an active promotional campaign with remaining redemptions.

Flow: The attendee scans the Campaign QR Code, opens the event page with the promo code pre-applied, selects tickets, reviews the discount, and completes checkout.

Postcondition: The discount is recorded on the order and the sale is attributed to the correct campaign.

UC11: Organizer Reviews Basic Analytics

Actor: Organizer

Precondition: The organizer manages an event with ticket, order, campaign, or check-in activity.

Flow: The organizer opens the event dashboard, selects a date range or ticket type, and reviews sales progress, revenue, campaign performance, and attendance.

Postcondition: The organizer sees metrics calculated from BiletFlow records without collecting additional attendee information.

UC12: Organizer Reviews and Reuses Event History

Actor: Organizer

Precondition: The organizer manages at least one completed or cancelled event.

Flow: The organizer opens the past-events view, reviews the event activity timeline and retained operational records, then duplicates the event into a new draft.

Postcondition: Historical records remain unchanged and the new draft contains reusable event configuration without copying old transactions or attendees.

6. Core Data Entities

- User
- Organizer Profile
- Event
- Venue
- Venue Section
- Row
- Seat
- Seat Hold
- Ticket Type
- Order
- Order Item
- Ticket
- Attendee
- Payment
- Refund
- Payout Account
- Check-In Record
- Staff Assignment
- Notification
- Support Case
- Support Message
- Promotional Campaign
- Promo Code
- Promo Redemption
- Audit Log

7. Non-Functional Requirements

- The platform should support modern desktop and mobile browsers.
- The ticket-verification app shall support current iOS and Android versions selected for the MVP.
- The customer-facing interface shall initially support Kazakh and Russian, with English available as an additional locale.
- Prices, fees, refunds, and payouts shall use KZT for the initial Kazakhstan release.
- Sensitive information shall be encrypted in transit and at rest.
- Passwords shall be stored using secure password hashing.
- Payment-card data shall not be stored directly by the platform.
- Organizer and administrator actions shall be auditable.
- Event pages should meet WCAG 2.1 AA accessibility guidelines.
- The checkout process should remain usable under temporary increases in event traffic.

- Seat availability shall update promptly, and checkout shall use an atomic reservation process to prevent double-selling a seat.
- The interactive seat map shall be usable on desktop and mobile screens and provide accessible labels in addition to color indicators.
- QR validation should normally complete within two seconds with an internet connection.
- Printable tickets shall produce legible text and reliably scannable QR codes on standard A4 paper in color or grayscale.
- Calendar exports shall preserve the event's configured time zone.
- Support-case access shall be enforced by role and by relationship to the relevant event, order, or ticket.
- Promo-code validation and redemption limits shall be enforced atomically on the server.
- Campaign QR links shall not contain trusted price or discount values and shall never be accepted by ticket-verification endpoints.
- Analytics queries and chart generation shall not block or delay ticket selection, checkout, payment confirmation, or ticket issuance.
- Organizer history and audit data shall be protected by the same event-level authorization rules as the underlying operational records.
- The architecture should support additional currencies and languages in future releases.
- Regular backups and recovery procedures shall be established.

8. MVP Scope

Required MVP Features

- User registration and authentication
- Organizer profiles
- Event creation and publication
- Free and paid ticket types
- Simulated or sandboxed paid-sales activation
- Simulated or sandboxed KZT checkout
- QR-code ticket generation
- Downloadable and printable PDF tickets
- Email confirmations
- Attendee and order management
- Dedicated iOS and Android ticket-verification app for Event Admins
- Basic cancellations and refunds
- Administrative moderation
- Basic organizer analytics for capacity, sales over time, revenue, ticket types, campaigns, and attendance
- Organizer event history with past/completed/cancelled views, event duplication, and an activity timeline
- Asynchronous support chat with case status and contextual event/order/ticket links

- Promo codes with special Campaign QR Codes and basic campaign sales reporting
- Docker-based local or demonstration deployment

Bonus and Stretch Features

- One predefined assigned-seating layout with an interactive seat-selection UI
- Calendar export through .ics files and common calendar links
- Advanced organizer analytics using GA4 traffic/funnel data combined with authoritative BiletFlow order data
- Offline ticket-verification synchronization
- Support-message file attachments or enhanced real-time delivery

Excluded from the Initial MVP

- Real-money organizer payouts and production financial processing
- Production KYC/KYB or identity-verification workflows
- Visual venue-layout designer and arbitrary venue imports
- Production-grade offline scanner synchronization
- Advanced refund and dispute workflows
- App Store and Google Play publication
- Ticket resale or transfer marketplace
- Native attendee and organizer mobile applications
- Affiliate marketing
- Recurring events
- Advanced marketing automation
- On-site hardware integrations
- Multiple organizer payout splits
- Detailed accounting and tax automation
- Full field-by-field version history, rollback, and production compliance archival

9. Suggested Technology Stack

The frontend and mobile technologies are recommended for consistency, while the team may choose the backend language that best matches its experience. The academic MVP shall use one primary backend implementation; creating the same services in multiple languages or introducing unnecessary microservices is outside the term scope.

- Database: PostgreSQL or MongoDB with Mongoose. PostgreSQL may be paired with Node.js, Python, Go, or Java; selecting Node.js does not require MongoDB.

- Backend Option 1 - JavaScript/TypeScript: Node.js with Express or a comparable framework; this produces a MERN-style stack when combined with MongoDB and React
- Backend Option 2 - Python: FastAPI or Django
- Backend Option 3 - Go: Go with a lightweight HTTP framework or the standard HTTP library
- Backend Option 4 - Java: Spring Boot
- Web Application: React; Next.js may be used when server-side rendering or search visibility is needed
- Seat Map UI: React with SVG or Canvas rendering, one predefined layout, and server-validated seat holds
- Mobile Ticket-Verification App: React Native, optionally with Expo
- Frontend Language: TypeScript is recommended across the web application and React Native app; it may also be shared with the backend when Node.js is selected
- Styling: Tailwind CSS
- Payments: Provider sandbox or internal simulation using KZT; production provider integration is deferred
- Email: Transactional email provider
- File Storage: S3-compatible object storage
- Printable Tickets: Server-generated PDF based on the canonical ticket record
- Support Chat: REST API with periodic refresh or lightweight polling; real-time sockets are optional and not required
- Campaign QR Codes: Server-generated QR images containing signed or opaque promotional links
- Analytics: Database aggregation for exact sales/redemption data and optional GA4 ecommerce/campaign events for traffic attribution
- Deployment: Docker containers
- Design: Figma
- Testing: Automated API, interface, payment, and QR-validation tests

The backend language and database shall be selected during the initial architecture phase and documented with a short rationale covering team familiarity, development speed, data-model requirements, and deployment complexity.

10. Assumptions and Dependencies

- The team can access a suitable payment sandbox or implement a clearly labelled payment simulation.
- The initial release will operate in Kazakhstan and use KZT as its settlement currency.
- Payment, privacy, refund, and organizer-verification rules will be reviewed for compliance with applicable Kazakhstan requirements before launch.
- Organizers are responsible for event legality, descriptions, and refund policies.
- Attendees have access to email or an online account for ticket delivery.

- Event Admins have supported iOS or Android devices with working cameras.
- Event venues provide internet access for real-time check-in; offline check-in is outside the academic MVP.
- AI-assisted tools may be used during development; this document does not prescribe a particular tool or AI-assisted workflow.
- Final production fees, supported payment methods, payment provider, and payout timing remain future decisions.

11. Initial Success Criteria

The MVP will be considered successful when:

- An organizer can publish a free event and distribute valid tickets.
- An organizer can complete the simulated or sandboxed paid-sales activation and receive demonstration orders.
- An attendee can complete checkout and receive a QR-code ticket.
- An attendee can download and print a ticket that remains scannable at check-in.
- An Event Admin can use the mobile app to validate tickets and prevent duplicate entry.
- An attendee can open a contextual support case and exchange asynchronous messages with authorized organizer staff.
- An attendee can scan a Campaign QR Code, receive a valid discount, and complete an order attributed to the campaign.
- An admission scanner rejects Campaign QR Codes while continuing to accept valid ticket QR codes.
- Organizers can view accurate ticket, attendance, payment, and refund records.
- Organizers can view basic sales, capacity, campaign, and attendance analytics without adding fields to the attendee checkout.
- Organizers can review past and cancelled events, inspect an authorized activity timeline, and duplicate a past event without copying historical transactions.
- Administrators can suspend a suspicious event and stop further sales.

Bonus features are evaluated separately and are not required to satisfy the base MVP success criteria.

12. Key Decisions for the Next Draft

- Final project and product name
- Final backend selection among Node.js, Python, Go, and Java, together with MongoDB or PostgreSQL
- Activation fee amount and refundability
- Payment sandbox or simulation method for the academic MVP

- Whether attendees must create accounts
- Whether processing fees are paid by organizers or attendees
- Payout delay and reserve policy
- Organizer verification requirements
- Event cancellation and refund rules
- Predefined seating layout used for the demonstration
- Whether support cases allow file attachments in the academic MVP
- Initial promo-code discount types, redemption limits, and campaign reporting fields
- Default date ranges and chart formats for the basic organizer dashboard
- Academic retention period for event history and audit entries

13. Term Feasibility and Delivery Plan

13.1 Feasibility Target

The project is planned as a polished academic prototype developed by a team of five over 10–12 weeks. The team may use AI-assisted development tools. The term deliverable is not intended to be a production financial platform.

The MVP is feasible only if the team maintains the scope boundaries in Section 8. Real payments, real payouts, production identity verification, arbitrary venue-map creation, offline scanner synchronization, and application-store publication are not term requirements.

13.2 Suggested Team Ownership

Each area shall have one primary owner, with peer review from at least one other team member:

1. Architecture, integration, authentication, and deployment
2. Backend event, ticket, order, and seat-reservation services
3. Organizer and attendee web interfaces
4. Admin interface, promotional campaigns, basic reporting, organizer history, calendar bonus, and printable tickets
5. React Native verification app, support workflow, assigned-seating bonus, automated testing, and release coordination

Ownership does not prevent collaboration. Shared API contracts and data models should be agreed before parallel implementation begins.

13.3 Suggested 12-Week Schedule

- Weeks 1–2: select one backend stack, confirm scope, prepare wireframes, define API contracts and data model, and configure repositories and automated checks
- Weeks 3–4: implement authentication, roles, organizer profiles, and event creation
- Weeks 5–6: implement ticket types, inventory, sandbox checkout, orders, and QR generation
- Week 7: implement printable PDF tickets, promo codes, and Campaign QR Codes; add calendar export only if the core flow is on schedule
- Week 8: implement the React Native online verification app and asynchronous support workflow
- Week 9: implement basic administration, organizer history, and the required analytics dashboard; add assigned seating or advanced GA4 analytics only if the core flow is stable
- Week 10: integrate features, resolve cross-application defects, and complete core-flow tests
- Weeks 11–12: stabilize, improve usability, prepare documentation, rehearse the demonstration, and retain contingency time

For a 10-week term, Weeks 7–9 should overlap where team capacity permits, while Week 10 remains reserved for integration and stabilization.

13.4 Scope and Quality Controls

- Core end-to-end flow takes priority: create event, distribute a campaign QR, apply a promo code, select or reserve a ticket, complete simulated checkout, issue the ticket, request support, and verify entry.
- Feature development should stop after Week 9 unless a missing feature blocks the core demonstration.
- Each feature is complete only when integrated, reviewed, tested, and documented.
- Bonus features shall not be started at the expense of an incomplete required flow.
- If the project falls behind, remove assigned seating, calendar export, advanced analytics, localization polish, and refund simulation before cutting the core ticketing flow.